DC Energy Meter

➢ Requirements:

A device to monitor Voltage and Current bidirectionally and store the data in a SD Card. Minimum specs: 100V 50~100A.

➢ Initial Solution:

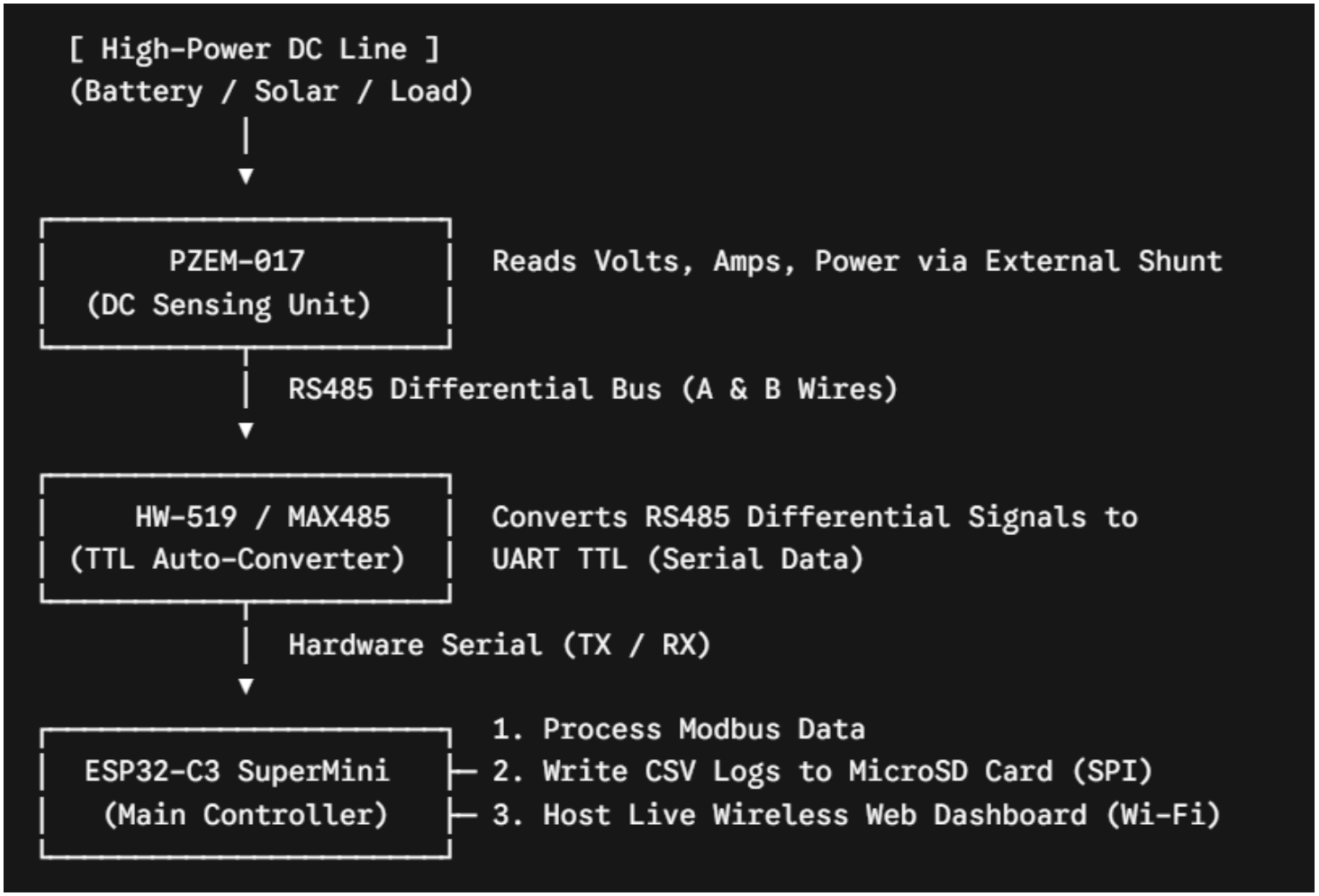


Peacefair PZEM-017 with 100A shunt resistor

- **0.05 - 300V**
- **0.02 – 300A**
- **1% accuracy**

This system acts as a standalone DC energy logger and wireless dashboard. It measures high-power DC voltage and heavy current via an external 100A/75mV brass shunt using a PZEM-017 module. The PZEM-017 transmits measurement data via Modbus RTU over an RS485 bus, which an auto-flow converter translates into 3.3V serial signals. The ESP32-C3 SuperMini processes these metrics, simultaneously writing time-stamped .CSV logs

onto a MicroSD card and hosting a local Wi-Fi web server to display real-time graphical graphs on a laptop or phone.



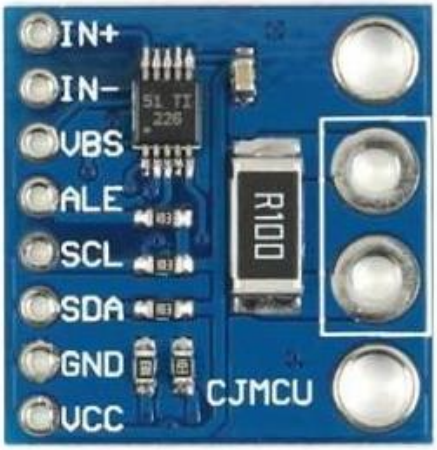
- **Problem:**

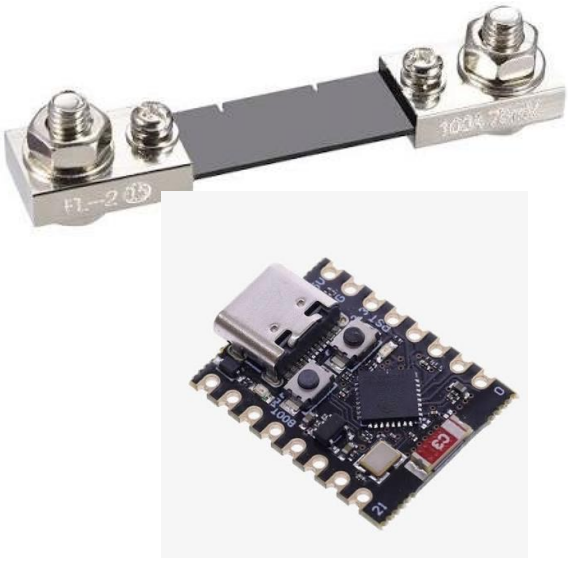
  Imported from China by robu.in and all variants are sold out.

  Kind-of expensive (900~5000+)

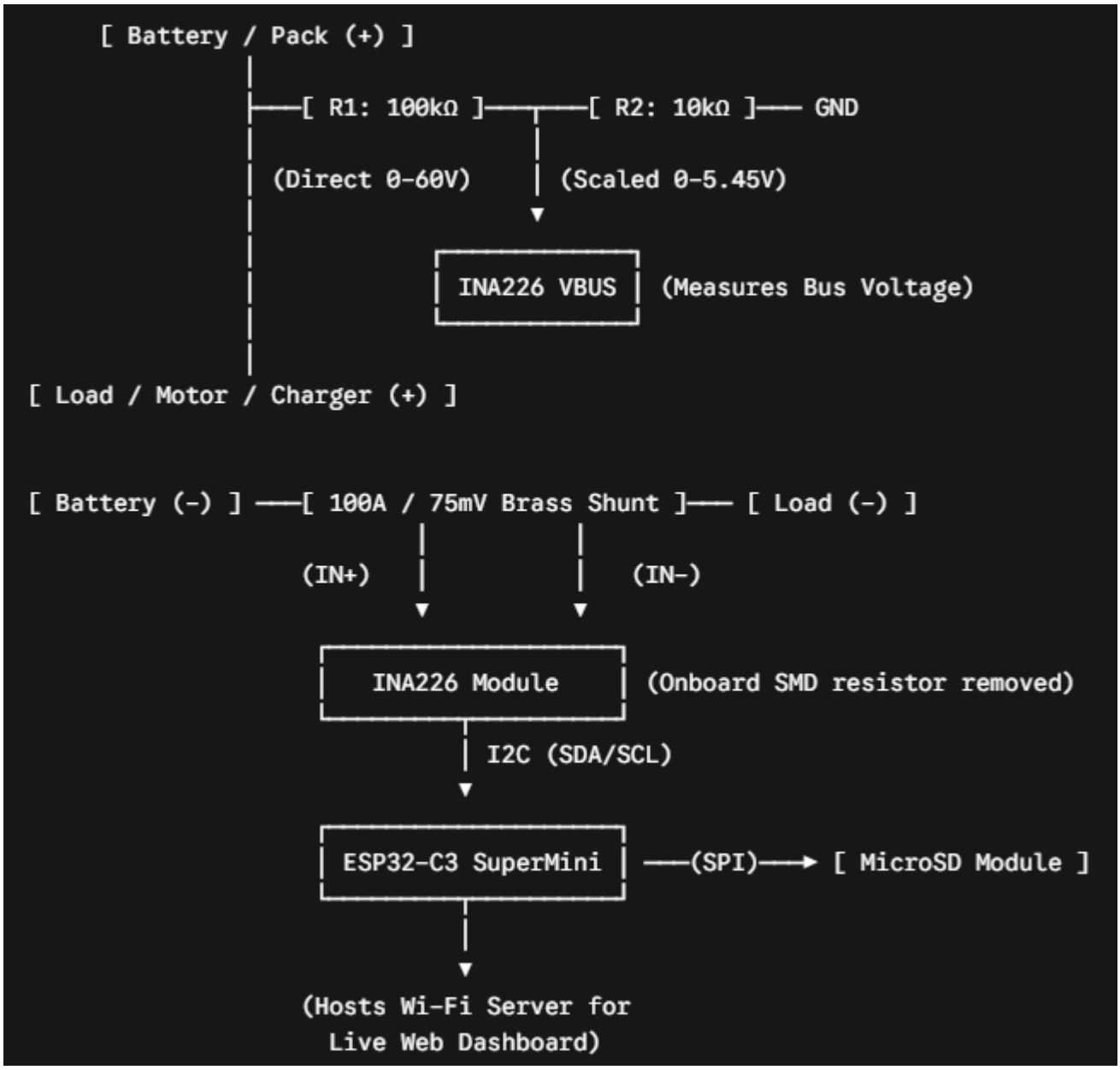
  Requires TTL to RS485 converter to communicate through UART

➢ Proposed Solution:





This system serves as a standalone DC energy logger and wireless dashboard. High system current passes through an external 100A/75mV brass shunt, generating a millivolt drop measured by an INA226 module (with its onboard resistor removed). A 0.1% precision voltage divider scales pack voltages (0--60V) down to safe levels for the INA226. The INA226 sends 16-bit measurements over I^2C to an ESP32-C3 SuperMini, which scales the values back, writes timestamped .CSV logs to a MicroSD card via SPI, and hosts a Wi-Fi server displaying real-time graphical data on a laptop or phone.


[Battery / Pack (+)]
[R1: 100kΩ]
[R2: 10kΩ]
GND
(Direct 0-60V)
(Scaled 0-5.45V)
INA226 VBUS
(Measures Bus Voltage)
[Load / Motor / Charger (+)]
[Battery (-)]
[100A / 75mV Brass Shunt]
[Load (-)]
(IN+)
(IN-)
INA226 Module
(Onboard SMD resistor removed)
I2C (SDA/SCL)
ESP32-C3 SuperMini
(SPI)
[MicroSD Module]
(Hosts Wi-Fi Server for
Live Web Dashboard)